# Can a smaller giveaway make a better product launch?

## An agent-based decision experiment for a hypothetical campus bottle seller

**300**
individual customers

**60**
campaign periods

**36**
tested policy cells

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Scope: MODEL_SPEC.md; 36 cells include the exploratory three-sample extension.

# The seller chooses the launch. Customers decide to buy.

## CHOOSE

Price: Rs 400 / 500 / 600

Free bottles: 3 or 12

Who receives them?

## SIMULATE

Customers talk to contacts

Opinions and awareness change

Eligible individuals buy once

## EVALUATE

Paid buyers, not free owners

Revenue and all bottle costs

Profit and uncertainty

Discussion-contact policy is an additional scenario: random or similar-opinion replacements.

Boundary: one product; one purchase/customer; no inventory shortage or fixed overhead. MODEL_SPEC.md

# Synthetic inputs are specified, not fitted to campus data.

## 300 people, 1,200 ties

Three known groups; stronger initial preference for within-group contacts.

## Budgets and value

Belief Beta(2,2)

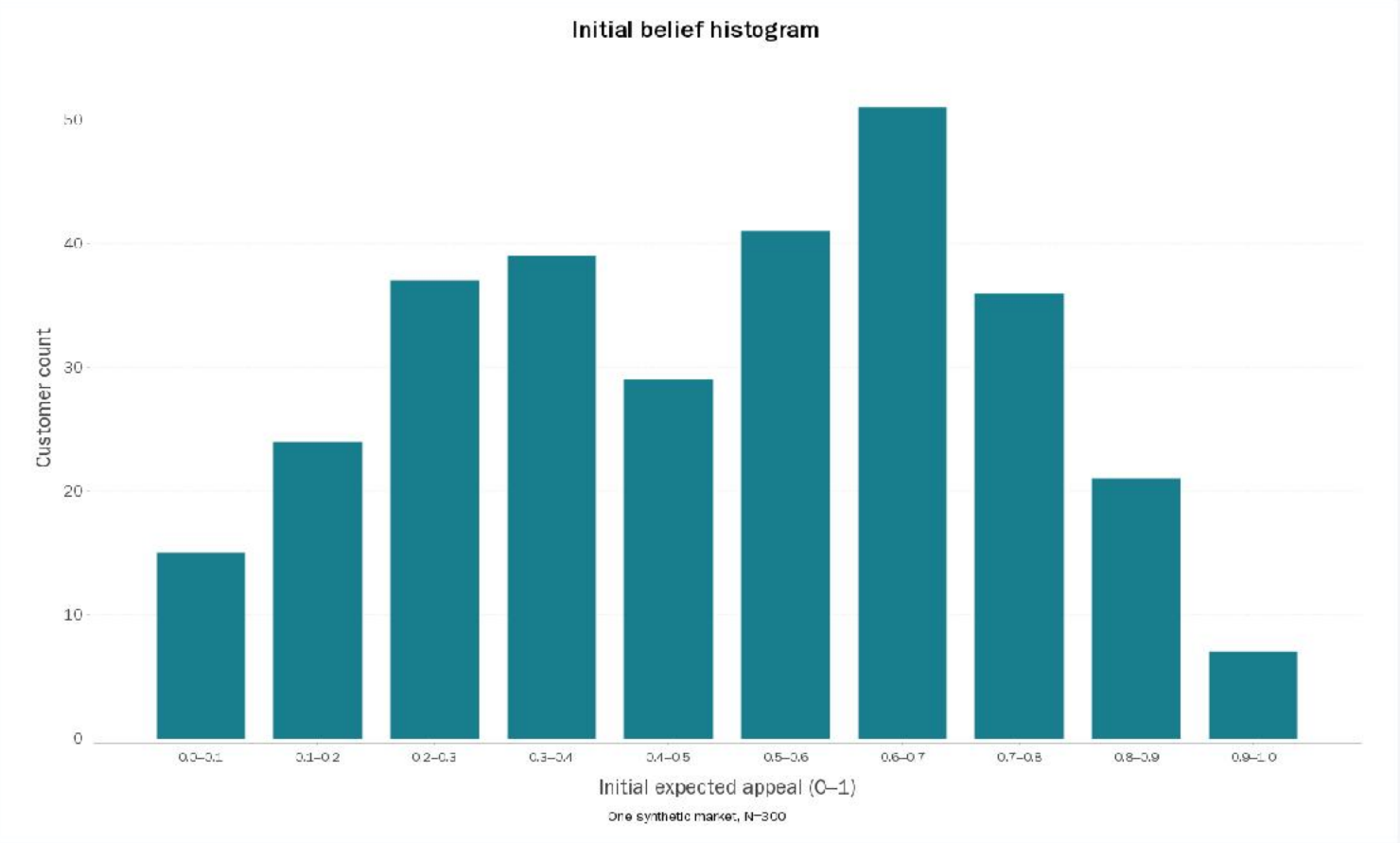
Value scale Rs 500–1,000

Affordability Rs 400–1,000

## Example market, rep 1000

Belief mean 0.490; SD 0.235.

Budget mean Rs 710.93.




data/customers_rep1000.csv; sim/analysis/input_statistics.csv. Histogram is one synthetic market.

# A conversation changes beliefs. A purchase needs value AND budget.

| | | |
|---|---|---|
| 1 | INFORM | An aware contact shares launch news with probability 0.25. |
| 2 | DISCUSS | Close beliefs compromise: 0.40 and 0.50 become 0.43 and 0.47. |
| 3 | BUY ONCE | Aware + can afford + perceived value ≥ price. Purchases at period end. |
| 4 | EXPERIENCE | New owners receive personal feedback. Beliefs and later conversations can change. |

Rules: MODEL_SPEC.md. d=0.2; mu=0.3; phi=0.05. Worked opinion example is illustrative.

# Established mechanisms; explicit new assumptions; checked code.

## LITERATURE

Bounded confidence and adaptive ties

Promotional seeding and diffusion

No claim to invent these mechanisms

## OUR ASSUMPTIONS

Price/value/budget threshold

Experience resets and cost accounting

Synthetic graph and attributes

## EVIDENCE

16 product checks passed

First AnyLogic default matches Java

Revised code: actual API compile passes

Deffuant et al. 2000; Kozma & Barrat 2008; Delre et al. 2007/2010; Kempe et al. 2003. See LITERATURE_NOTES.md and VALIDATION_REPORT.md.

# Compare policies on the same market; confirm on fresh markets.

## PILOT

30 separate markets

18 policies

Choose a provisional policy

## MAIN + FOLLOW-UP

100 main markets

18 + 18 policies

3,600 evaluations, paired

## CONFIRMATION

100 fresh markets

4 fixed comparisons

No search for a new winner

95% intervals: 10,000 bootstrap resamples of markets.

Customers, periods and reused policies are not independent replications.

sim/results/*manifest.json; sim/analysis/confirmation_paired.csv. Exploratory follow-up explicitly distinguished from initial design.

# Almost everyone hears. Far fewer buy at the high price.

## News is not conversion

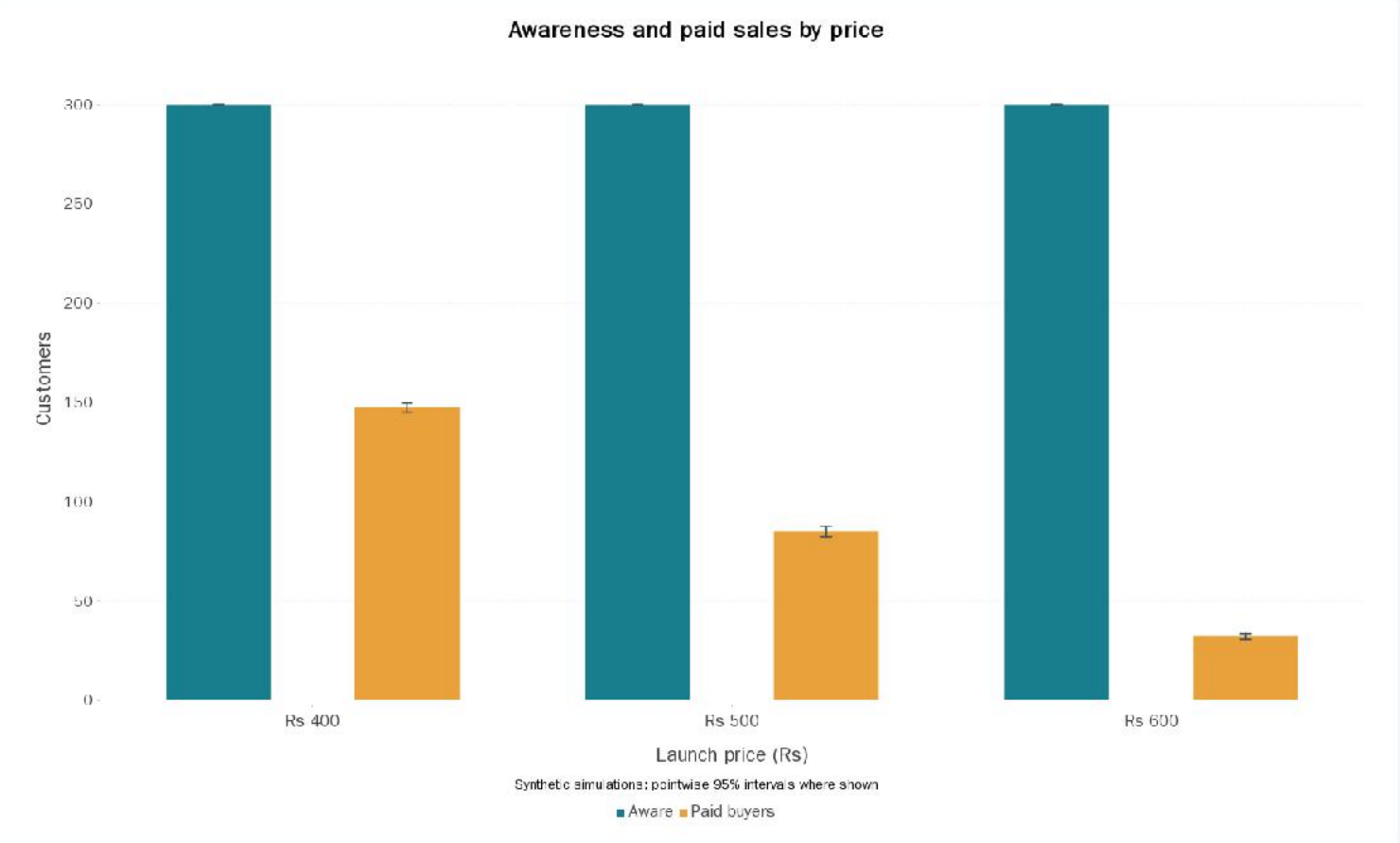
At Rs 600, awareness is nearly 300; mean paid buyers are 32.21.

## Margin is not profit

Profit means: Rs 19,100 / 18,213 / 8,274 at the three prices.

## Same campaign setup

12 samples; degree targeting; random contact replacement.




sim/analysis/policy_summary.csv; 100 main markets/cell; error bars are pointwise 95% intervals for means.

# Nine fewer free bottles can increase profit while reducing total owners.

## + Rs 2,854.50

Fresh-market paired profit gain

95% CI: Rs 2,698.50–3,006

## Why it happens

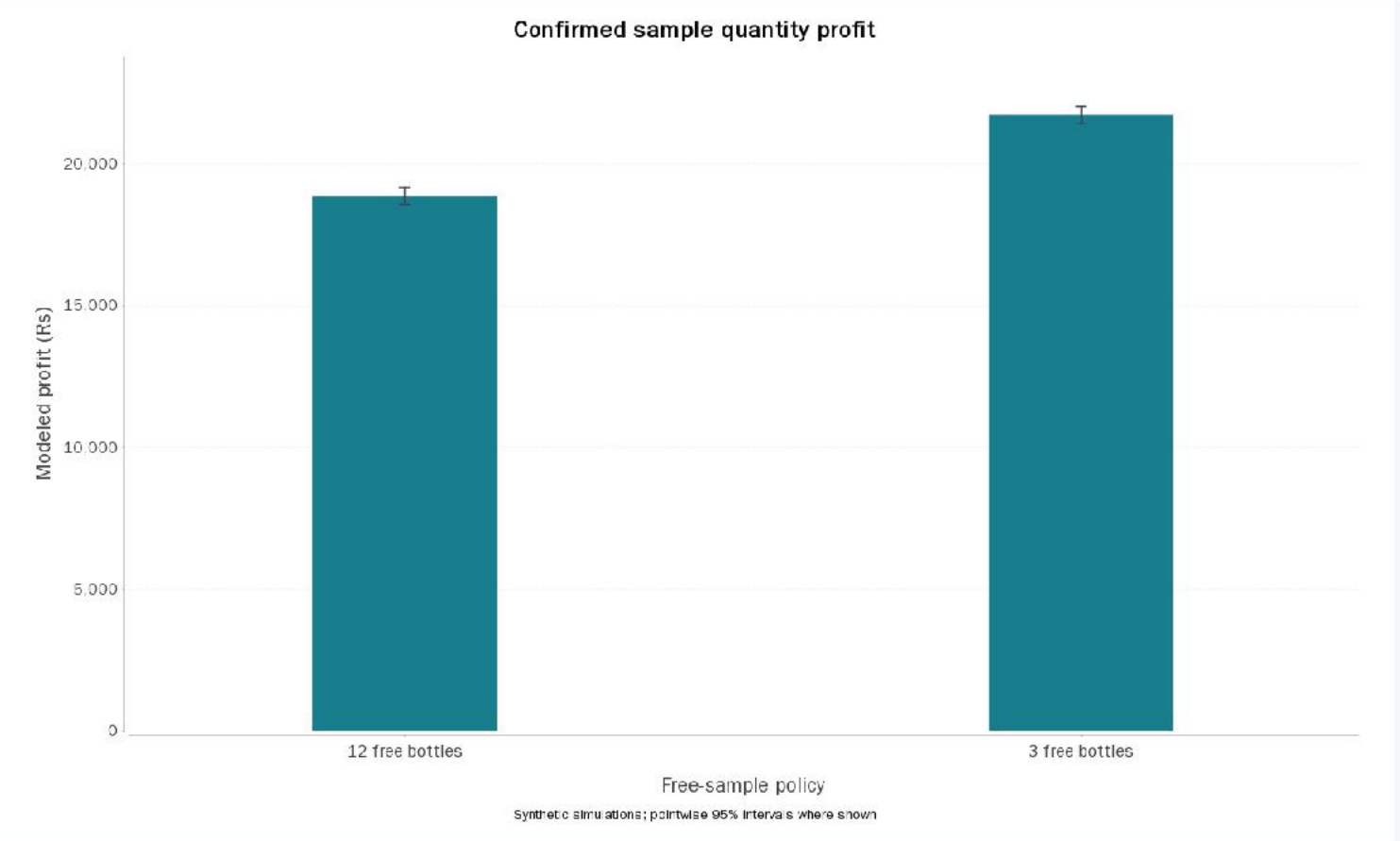
Rs 2,250 less sample cost,

plus margin on about 4 extra sales.

## Different objectives differ

Paid sales rise; total owners fall.

Three vs twelve is not a global quantity optimum.




sim/analysis/confirmation_summary.csv and confirmation_paired.csv; 100 fresh paired markets; price Rs 400, degree, h=0.

# The sample saving is clear. The influencer premium is uncertain.

## Degree over random

+ Rs 31.50

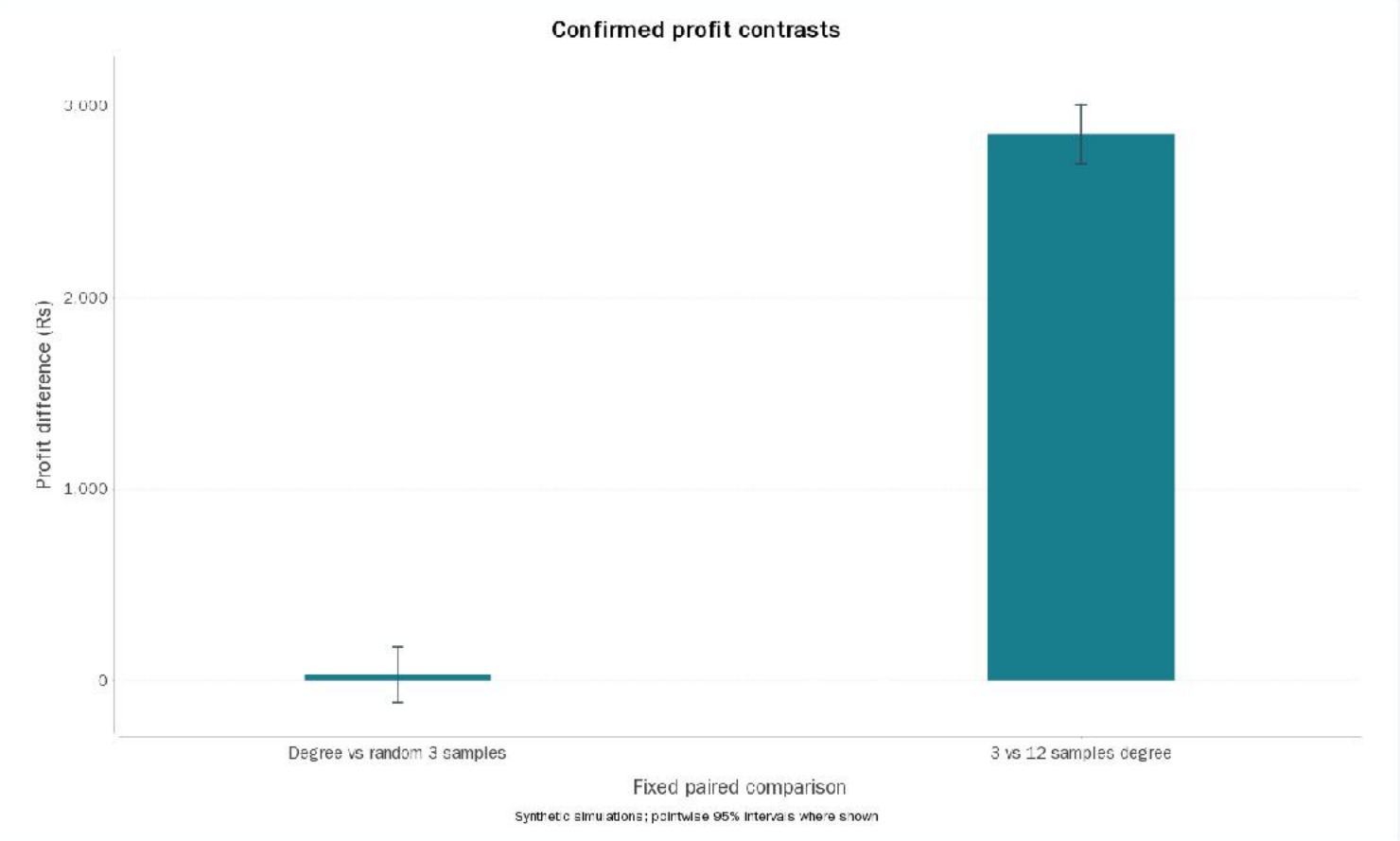
95% CI: −112.50 to +177

## Avoid expensive targeting

We have not established a large final-profit benefit from degree information.

## Timing can still matter

Strong-group stress: distributed seeding helps early coverage, not clearly final profit.




confirmation_paired.csv; stress_paired.csv. Pointwise intervals; lack of a clear difference is not equivalence.

# A surprise exposes a risky assumption: very positive opinions can be ignored.

## At narrow confidence

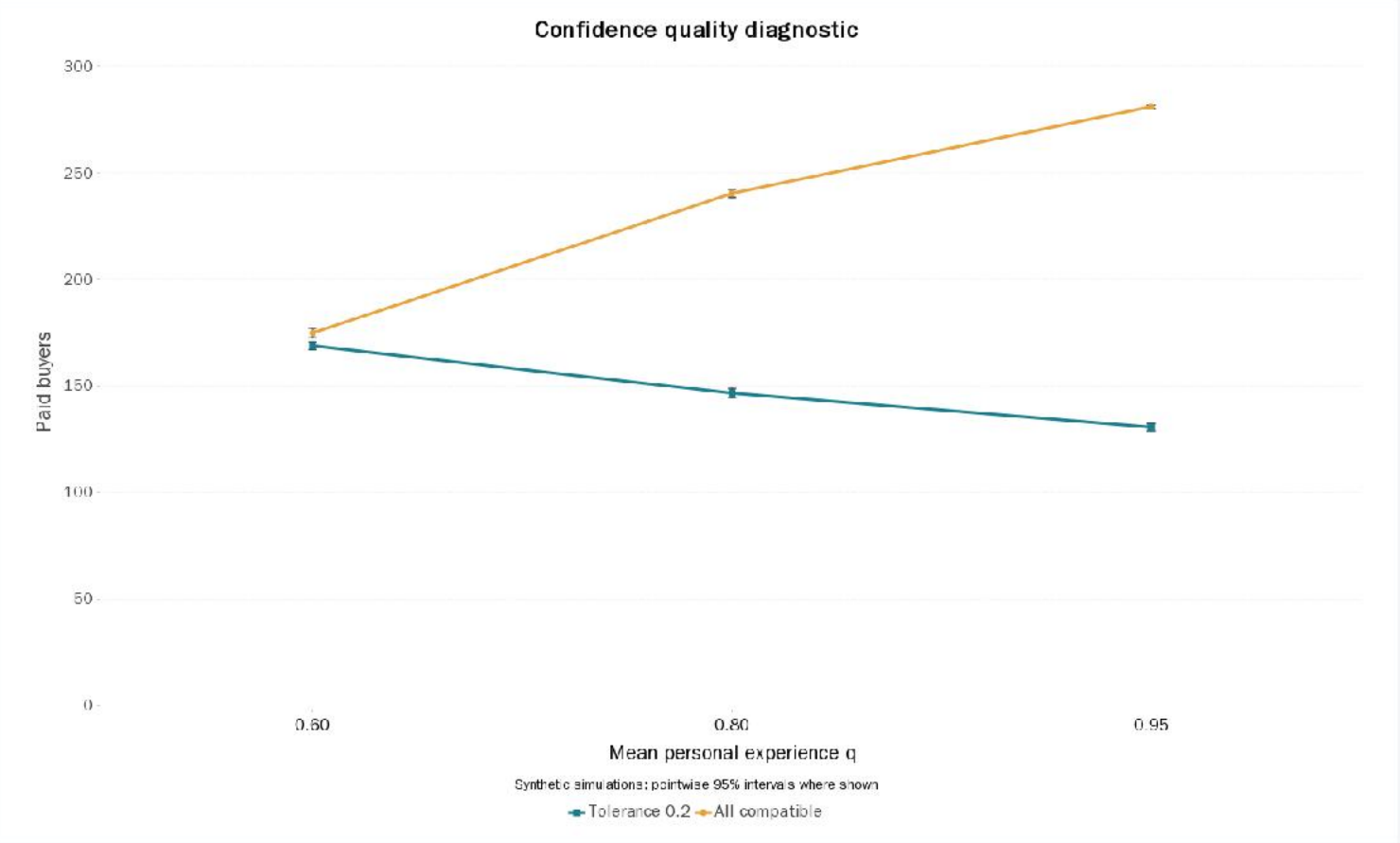
Higher experience mean can reduce simulated sales.

## Remove the gate

With all opinions compatible, the quality effect becomes positive.

## Interpretation

Validate or soften the attention rule. Do not recommend worse products.




quality_diagnostic_summary.csv; quality_paired.csv. Exploratory 100 paired markets/setting; fixed graph, Rs 400, 12 samples, degree.

# Live: follow one conversation, then check the campaign accounts.

## WATCH

Agent appearance: beliefs and ownership

Selected link: a recent meeting

Awareness versus paid sales

## EXPLAIN

Close opinions move together

Distant opinions may change contacts

Buying requires budget and value

## CHECK

Default: 148 paid + 12 free

Revenue Rs 59,200 − cost Rs 40,000

Profit Rs 19,200

Run the engine-tested model. Defaults: policyIndex=6, rep=1000. Keep the grid model as fallback.

Approved grid default has real screenshot/CSV evidence. LiveNetwork API/wiring pass; its actual engine run is tracked separately.

# Start small. Revisit price with cost. Validate the behavior before deployment.

01 Baseline candidate: Rs 400, three samples. Random targeting remains a practical alternative.

02 Budget is a ceiling, not a spending target. Cost changes can change the preferred price.

03 Synthetic assumptions drive outcomes. The quality reversal is a warning to validate behavior.

Best among tested candidates; no claim of a global optimum or actual-campus sales forecast.

confirmation_paired.csv; all_cost_revaluation.csv; PROJECT_REPORT.md. AI-assisted research, code and drafting.

# Profit versus price under the original equal-sample design.




all_policy_summary.csv. 12 samples, h=0; 100 markets; pointwise mean intervals.

# The preferred price depends on unit cost.




all_cost_revaluation.csv. 3 samples, degree, h=0; cost-only re-accounting holds customer behavior fixed.

# What we did not establish.

- No real demand calibration or empirical sales validation.
- No global optimum: only three prices and two sample quantities.
- No costly network-data collection, competition, inventory or fixed overhead.
- Hard confidence, independent attributes and informed-nonbuyer sharing are assumptions.
- Finite campaign outcomes are not equilibrium claims.
- Pointwise intervals do not protect every comparison simultaneously.
- Actual engine evidence must be kept separate from reference and Java runs.

OPEN_ISSUES.md; VALIDATION_REPORT.md; MODEL_SPEC.md.

# Sources and the boundary of our contribution.

Deffuant et al. (2000), Mixing beliefs among interacting agents.

Kozma & Barrat (2008), PRE 77, 016102. doi:10.1103/PhysRevE.77.016102

Delre et al. (2007), JBR 60, 826–835. doi:10.1016/j.jbusres.2007.02.002

Delre et al. (2010), JPIM 27, 267–282. doi:10.1111/j.1540-5885.2010.00714.x

Kempe, Kleinberg & Tardos (2003), Maximizing the Spread of Influence.

Replication-informed extension; seeding and homophilic rewiring are established.

Full reading limits and claim checks are preserved in LITERATURE_NOTES.md.

Direct source links and remaining extraction limits are in the literature notes; no research-priority claim.